THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 60°	Complementary angles add up to 90°.
2	**(a)** 130°	Supplementary angles add up to 180°.
3	**(b)** 40°	Vertically opposite angles are equal.
4	**(c)** 120°	A linear pair adds up to 180°.
5	**(c)** parallel lines	Recall the definition.
6	**(d)** 55°	Corresponding angles are equal when the lines are parallel.
7	**(b)** 48°	Alternate interior angles are equal when the lines are parallel.
8	**(a)** 118°	Interior angles on the same side add up to 180°.
9	**(b)** 25°	Complementary angles add up to 90°.
10	**(d)** 65°	Supplementary angles add up to 180°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Complement $= 90^\circ - 35^\circ = 55^\circ$ (1); Supplement $= 180^\circ - 110^\circ = 70^\circ$ (1).	1+1
12	Vertically opposite angle $= 65^\circ$ (1); The other two angles form a linear pair with it: $180^\circ - 65^\circ = 115^\circ$ each (1).	1+1
13	Vertically opposite angle $= 70^\circ$ (1); Linear pair: $180^\circ - 70^\circ = 110^\circ$ (1).	1+1
14	Complement $= 90^\circ - 62^\circ = 28^\circ$ (1); Supplement $= 180^\circ - 75^\circ = 105^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 180^\circ$ and the number of parts $= 3$ (1); One part $= 60^\circ$ (1); The angles are 60° and 120° (1).	1+1+1
16	$x = 180 - 110 = 70$ (1); The angle 70° is acute (2).	1+2
17	(i) Alternate interior angles are equal: 52° (1); (ii) Interior angles on the same side add up to 180°: 128° (2).	1+2
	OR: Sum $= 180^\circ$ and the number of parts $= 6$ (1); One part $= 30^\circ$ (1); The angles are 30° and 150° (1).	1+1+1
18	$x = 180 - 135 = 45$ (1); The angle 45° is acute (2).	1+2
	OR: (i) Alternate interior angles are equal: 66° (1); (ii) Interior angles on the same side add up to 180°: 114° (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Vertically opposite angle $= 35^\circ$ (1); The other two angles are 145° each (linear pair) (1); Sum $= 35 + 35 + 145 + 145 = 360$ (1); The four angles around the point add up to 360° (2).	1+1+1+2
	OR: Let the complement be y; the angle is $2y$ and $2y + y = 90$ (1); $y = 30$ (2); The angle is 60° and its complement is 30° (2).	1+2+2
20	Vertically opposite angle $= 50^\circ$ (1); The other two angles are 130° each (linear pair) (1); Sum $= 50 + 50 + 130 + 130 = 360$ (1); The four angles around the point add up to 360° (2).	1+1+1+2
	OR: Let the complement be y; the angle is $4y$ and $4y + y = 90$ (1); $y = 18$ (2); The angle is 72° and its complement is 18° (2).	1+2+2

Mistakes to avoid:

- Mixing up complementary (90°) and supplementary (180°) angles.
- Saying vertically opposite angles are supplementary; they are equal.
- Saying corresponding or alternate interior angles are supplementary; same-side interior angles are the supplementary pair.
- Using the corresponding angle result when the lines are not known to be parallel.
- Forgetting that the four angles around a point add up to 360°, not 180°.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 75°	Vertically opposite angles are equal.
2	**(b)** 80°	A linear pair adds up to 180°.
3	**(a)** transversal	Recall the definition.
4	**(c)** 82°	Corresponding angles are equal when the lines are parallel.
5	**(b)** 73°	Alternate interior angles are equal when the lines are parallel.
6	**(a)** 95°	Interior angles on the same side add up to 180°.
7	**(c)** 42°	Complementary angles add up to 90°.
8	**(c)** 102°	Supplementary angles add up to 180°.
9	**(a)** 110°	Vertically opposite angles are equal.
10	**(c)** 55°	A linear pair adds up to 180°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Vertically opposite angle $= 110^\circ$ (1); The other two angles form a linear pair with it: $180^\circ - 110^\circ = 70^\circ$ each (1).	1+1
12	Vertically opposite angle $= 105^\circ$ (1); Linear pair: $180^\circ - 105^\circ = 75^\circ$ (1).	1+1
13	Complement $= 90^\circ - 18^\circ = 72^\circ$ (1); Supplement $= 180^\circ - 140^\circ = 40^\circ$ (1).	1+1
14	Vertically opposite angle $= 40^\circ$ (1); The other two angles form a linear pair with it: $180^\circ - 40^\circ = 140^\circ$ each (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 180^\circ$ and the number of parts $= 5$ (1); One part $= 36^\circ$ (1); The angles are 72° and 108° (1).	1+1+1
16	$x = 180 - 150 = 30$ (1); The angle 30° is acute (2).	1+2
17	(i) Alternate interior angles are equal: 74° (1); (ii) Interior angles on the same side add up to 180°: 106° (2).	1+2
	OR: Sum $= 180^\circ$ and the number of parts $= 9$ (1); One part $= 20^\circ$ (1); The angles are 80° and 100° (1).	1+1+1
18	$x = 180 - 95 = 85$ (1); The angle 85° is acute (2).	1+2
	OR: (i) Alternate interior angles are equal: 88° (1); (ii) Interior angles on the same side add up to 180°: 92° (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Vertically opposite angle $= 65^\circ$ (1); The other two angles are 115° each (linear pair) (1); Sum $= 65 + 65 + 115 + 115 = 360$ (1); The four angles around the point add up to 360° (2).	1+1+1+2
	OR: Let the complement be y; the angle is $5y$ and $5y + y = 90$ (1); $y = 15$ (2); The angle is 75° and its complement is 15° (2).	1+2+2
20	Vertically opposite angle $= 80^\circ$ (1); The other two angles are 100° each (linear pair) (1); Sum $= 80 + 80 + 100 + 100 = 360$ (1); The four angles around the point add up to 360° (2).	1+1+1+2
	OR: Let the complement be y; the angle is $8y$ and $8y + y = 90$ (1); $y = 10$ (2); The angle is 80° and its complement is 10° (2).	1+2+2

Mistakes to avoid:

- Mixing up complementary (90°) and supplementary (180°) angles.
- Saying vertically opposite angles are supplementary; they are equal.
- Saying corresponding or alternate interior angles are supplementary; same-side interior angles are the supplementary pair.
- Using the corresponding angle result when the lines are not known to be parallel.
- Forgetting that the four angles around a point add up to 360°, not 180°.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** complementary angles	Recall the definition.
2	**(a)** 118°	Corresponding angles are equal when the lines are parallel.
3	**(c)** 99°	Alternate interior angles are equal when the lines are parallel.
4	**(d)** 76°	Interior angles on the same side add up to 180°.
5	**(a)** 18°	Complementary angles add up to 90°.
6	**(d)** 40°	Supplementary angles add up to 180°.
7	**(a)** 135°	Vertically opposite angles are equal.
8	**(d)** 145°	A linear pair adds up to 180°.
9	**(a)** perpendicular lines	Recall the definition.
10	**(b)** 67°	Corresponding angles are equal when the lines are parallel.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Vertically opposite angle $= 58^\circ$ (1); Linear pair: $180^\circ - 58^\circ = 122^\circ$ (1).	1+1
12	Complement $= 90^\circ - 47^\circ = 43^\circ$ (1); Supplement $= 180^\circ - 98^\circ = 82^\circ$ (1).	1+1
13	Vertically opposite angle $= 125^\circ$ (1); The other two angles form a linear pair with it: $180^\circ - 125^\circ = 55^\circ$ each (1).	1+1
14	Vertically opposite angle $= 132^\circ$ (1); Linear pair: $180^\circ - 132^\circ = 48^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 180^\circ$ and the number of parts $= 4$ (1); One part $= 45^\circ$ (1); The angles are 45° and 135° (1).	1+1+1
16	$x = 180 - 70 = 110$ (1); The angle 110° is obtuse (2).	1+2
17	(i) Alternate interior angles are equal: 103° (1); (ii) Interior angles on the same side add up to 180°: 77° (2).	1+2
	OR: Sum $= 180^\circ$ and the number of parts $= 9$ (1); One part $= 20^\circ$ (1); The angles are 40° and 140° (1).	1+1+1
18	$x = 180 - 160 = 20$ (1); The angle 20° is acute (2).	1+2
	OR: (i) Alternate interior angles are equal: 117° (1); (ii) Interior angles on the same side add up to 180°: 63° (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Vertically opposite angle $= 115^\circ$ (1); The other two angles are 65° each (linear pair) (1); Sum $= 115 + 115 + 65 + 65 = 360$ (1); The four angles around the point add up to 360° (2).	1+1+1+2
	OR: Let the complement be y; the angle is $14y$ and $14y + y = 90$ (1); $y = 6$ (2); The angle is 84° and its complement is 6° (2).	1+2+2
20	Vertically opposite angle $= 130^\circ$ (1); The other two angles are 50° each (linear pair) (1); Sum $= 130 + 130 + 50 + 50 = 360$ (1); The four angles around the point add up to 360° (2).	1+1+1+2
	OR: Let the complement be y; the angle is $17y$ and $17y + y = 90$ (1); $y = 5$ (2); The angle is 85° and its complement is 5° (2).	1+2+2

Mistakes to avoid:

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- Using the corresponding angle result when the lines are not known to be parallel.
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